

The Redshift Evolution of the MOND Acceleration Constant in the Causal Diamond Framework: A Falsifiable Prediction

Dat Nguyen

Independent Researcher

Dtnghuyen09@gmail.com | ORCID: 0009-0001-2514-7768

May 12, 2026

Abstract

In companion papers, we derived the MOND acceleration constant a_0 and the MOND interpolation function from the geometry and entropy structure of the cosmological causal diamond, with zero free parameters. Both quantities depend on the proper event horizon d_{eh} , which evolves with cosmic epoch. Here we compute the resulting prediction for $a_0(z)$ and compare it to four alternatives: constant MOND, the empirical formula from Paper 1 of this series, the Milgrom (2015) scaling $a_0 \propto cH(z)$, and Λ CDM simulations (Mayer et al. 2023). The causal diamond predicts a moderate increase: $a_0(z=2)/a_0(0) = 1.47$, distinct from constant MOND (1.00), the empirical formula (1.75), the Milgrom scaling (3.04), and Λ CDM simulations (2.75). All five predictions are cleanly separable at $z > 1$ and testable with next-generation surveys. The predicted evolution is robust to the entropy normalization ambiguity identified in the companion erratum, because the normalization cancels in the ratio $a_0(z)/a_0(0)$. This is the primary falsifiable prediction of the causal diamond program.

1. Introduction

The MOND acceleration constant $a_0 \approx 1.20 \times 10^{-10} \text{ m/s}^2$ [1] has been treated as a universal constant since Milgrom's original proposal [2]. Whether a_0 evolves with cosmic time is one of the most powerful observational tests that can distinguish between modified gravity and dark matter [3, 4], and between competing theoretical frameworks for MOND itself.

In companion papers [5, 6, 7], we showed that a_0 arises as the horizon-thermodynamic acceleration of the causal diamond waist in Λ CDM: $a_0 = c^2/(2\pi r_w)$, with 1.0% agreement and zero free parameters. The interpolation function was derived from area-law vs. volume-law entropy competition, with the volume-law term normalized to the proper event horizon d_{eh} [7]. Since d_{eh} depends on the cosmic epoch, the framework makes a specific, computable prediction for $a_0(z)$.

This paper computes that prediction, compares it to four alternatives, and identifies the redshift range where the models are observationally separable.

2. Five Predictions for $a_0(z)$

2.1 Model 1: Constant MOND

In standard MOND [2], a_0 is a fundamental constant: $a_0(z) = a_0(0)$. This is the null hypothesis.

2.2 Model 2: Causal diamond framework (this work)

In the entropy-balance derivation [6, 7], the transition from Newtonian to MOND gravity occurs where area-law entropy equals volume-law entropy. The volume-law entropy is bounded by the proper event horizon at the observation epoch. At redshift z , the proper event horizon is:

$$d_{\text{eh}}(z) = [1/(1+z)] \times c \int_{1/(1+z)}^{\infty} da' / [a'^2 H(a')] \quad (1)$$

where the comoving integral is converted to proper distance at epoch z by multiplication with the scale factor $a(z) = 1/(1+z)$. The MOND acceleration at epoch z is:

$$a_0(z) = 2c^2 / [3\pi d_{\text{eh}}(z)] \quad (2)$$

The use of the proper (rather than comoving) event horizon follows from the formulation of the entropy balance in Ref. [6]. The Bekenstein-Hawking area-law entropy $S_{\text{area}} = \pi r_{\text{pl}}^2 / l_{\text{pl}}^2$ is defined in terms of the proper area of the causal diamond cross-section [6, Sec. 2.1–2.2]. The volume-law entropy S_{vol} is similarly defined in terms of the proper volume [6, Eq. 3]. For the entropy balance to be self-consistent, the length scale normalizing the volume-law term must also be a proper distance. The proper event horizon $d_{\text{eh}}(z)$ is the physical causal boundary of the observer at epoch z , measured in the same units as the cross-section radius. The comoving event horizon, by contrast, mixes the physical distance at epoch z with the coordinate expansion between z and the future, and would be dimensionally inconsistent with the proper-distance entropy counting.

As z increases, the proper event horizon shrinks (the universe is physically smaller), reducing the dark energy entropy budget. The transition to MOND shifts to higher acceleration. The prediction is a moderate increase in a_0 with redshift.

2.3 Model 3: Empirical cosmological formula

In Paper 1 [8], we showed empirically that $a_0 = cH_0(\Omega_{\Lambda}/\Omega_m)^{1/6}/(2\pi)$. Extending to redshift z by replacing cosmological parameters with their epoch-dependent values:

$$a_0(z) = cH(z) [\Omega_{\Lambda}(z) / \Omega_m(z)]^{1/6} / (2\pi) \quad (3)$$

This predicts a 75% increase at $z = 2$.

2.4 Model 4: Milgrom (2015) scaling

Milgrom [4] considered the possibility that a_0 tracks the Hubble acceleration: $a_0(z) \propto cH(z)$. This gives $a_0(z)/a_0(0) = H(z)/H_0$, predicting a factor of 3.0 increase at $z = 2$. This is the simplest cosmological scaling and serves as a natural benchmark.

2.5 Model 5: Λ CDM hydrodynamical simulations

Mayer et al. [3] fitted MOND force laws to galaxies from the Magneticum Λ CDM simulation at multiple redshifts and found the best-fit a_0 increases by a factor of approximately 3 from $z = 0$ to $z = 2.3$. We approximate their results as $a_0(z)/a_0(0) \approx (1+z)^{0.92}$, noting that this is our parameterization of their discrete simulation data points, not a formula provided by the authors. In this model, the apparent MOND behavior is produced by evolving dark matter halos, not by modified gravity.

3. Numerical Results

We adopt Planck 2018 parameters [9]: $H_0 = 67.4$ km/s/Mpc, $\Omega_m = 0.315$, $\Omega_{\Lambda} = 0.685$, $\Omega_r = 9.15 \times 10^{-5}$. Table 1 reports $a_0(z)/a_0(0)$ for all five models.

z	Constant	Diamond	Empirical	Milgrom	Mayer
0.0	1.000	1.000	1.000	1.000	1.000
0.5	1.000	1.085	1.079	1.320	1.450
1.0	1.000	1.200	1.264	1.785	1.888

1.5	1.000	1.334	1.502	2.378	2.330
2.0	1.000	1.473	1.753	3.040	2.752
2.5	1.000	1.616	2.016	3.773	3.168
3.0	1.000	1.760	2.284	4.568	3.580

Table 1. Predicted $a_0(z)/a_0(0)$ for five models. The causal diamond (column 3) is distinct from all alternatives at $z > 1$. The Milgrom and Mayer predictions nearly coincide, reflecting their shared proportionality to the Hubble rate.

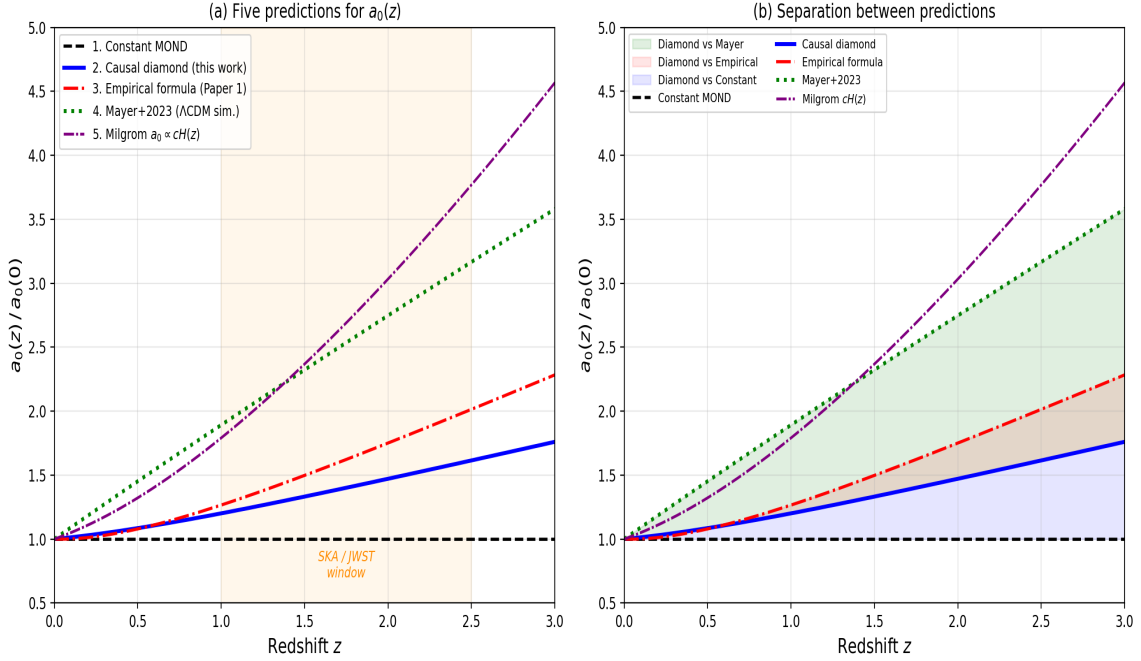


Figure 1. (a) Normalized $a_0(z)$ for five predictions. The causal diamond (blue) predicts the most moderate evolution, sitting well below the Milgrom/Mayer cluster at $z > 1$. The orange band marks the SKA/JWST window. (b) Shaded regions show the separation between the causal diamond and each alternative.

4. Distinguishability

At $z = 2$, the five predictions form three distinct groups:

Group A (no evolution): Constant MOND — $a_0/a_0(0) = 1.00$.

Group B (moderate evolution): Causal diamond (1.47) and empirical formula (1.75).

Group C (strong evolution): Mayer et al. (2.75) and Milgrom scaling (3.04).

Groups A, B, and C are separated by factors of ~ 1.5 – 2.0 , easily distinguishable with $\sim 20\%$ precision. Within Group B, the causal diamond and empirical formula differ by 16%, requiring $\sim 10\%$ precision to separate. Within Group C, Mayer and Milgrom differ by 10%. The SKA HI galaxy survey [10] will measure rotation curves for thousands of galaxies at $0.5 < z < 1.5$, and JWST is already producing resolved velocity fields at $z \sim 2$ [11].

5. Robustness

5.1 Independence from the normalization ambiguity

The companion erratum [7] identified an ambiguity in the volume-law entropy normalization that affects the absolute value of the interpolation parameter k . This ambiguity does not affect the present paper.

The predicted evolution is $a_0(z)/a_0(0) = d_{\text{eh}}(0)/d_{\text{eh}}(z)$, which is a ratio of proper event horizons. The normalization constant cancels identically. The evolution curve is therefore robust to the normalization choice.

5.2 The proper-distance convention

The entropy balance in Ref. [6] is formulated entirely in proper distances: the Bekenstein-Hawking area-law entropy uses the proper area of the causal diamond cross-section ($S_{\text{area}} = \pi r_{\text{proper}}^2 / l_{\text{Pl}}^2$), and the volume-law entropy uses the proper volume ($S_{\text{vol}} = (4\pi/3) r_{\text{proper}}^3 / (d_{\text{eh}} / l_{\text{Pl}})$). Self-consistency requires that the event horizon entering the normalization also be a proper distance. We therefore use the proper event horizon $d_{\text{eh}}(z) = a(z) \times d_{\text{eh,com}}(z)$.

We note that using the comoving event horizon instead would reverse the direction of the prediction: a_0 would *decrease* with redshift rather than increase, because the comoving event horizon grows at higher z (there is more comoving future ahead). The comoving choice would be dimensionally inconsistent with the proper-distance entropy counting of Ref. [6] and would predict that MOND effects weaken in the early universe, conflicting with the expectation that smaller physical horizons imply fewer available degrees of freedom. We report the proper-distance prediction as the one consistent with the framework's formulation, while noting that observational data will ultimately adjudicate.

6. Caveats

Several limitations apply. First, the Mayer et al. comparison uses our approximate parameterization $(1+z)^{0.92}$ of their discrete simulation data points. Their published results have scatter and error bars at each redshift snapshot; the smooth curve shown here is indicative, not exact.

Second, Eq. (2) assumes the entropy balance at each epoch can be evaluated using the instantaneous proper event horizon, treating the system as quasi-static. If the entropy balance responds with a time lag, the evolution would be smoother than predicted.

Third, the prediction is restricted to $z \leq 3$, the range where galaxy kinematics can be observed with current and near-future instruments. At $z > 5$, the proper event horizon becomes very small and the predicted a_0 grows to unphysical values. The framework's domain of validity is the epoch where dark energy is dynamically significant and galaxies exist to serve as probes.

Fourth, this paper makes a prediction *from* the causal diamond framework; it does not constitute evidence *for* it. The prediction becomes meaningful only when confronted with observational data.

7. Conclusion

The causal diamond framework predicts that a_0 increases with redshift as $a_0(z) = 2c^2/[3\pi d_{\text{eh}}(z)]$, where $d_{\text{eh}}(z)$ is the proper event horizon at epoch z . The predicted increase is 47% at $z = 2$ and 76% at $z = 3$.

This prediction is distinct from constant MOND (0%), the empirical formula (75% at $z = 2$), the Milgrom $cH(z)$ scaling (204%), and Λ CDM simulations (175%). It is the most moderate of the non-constant models, reflecting the physical picture that the proper event horizon shrinks slowly with redshift because dark energy remains dynamically present even when it does not dominate the expansion.

The evolution curve is robust to the entropy normalization ambiguity identified in Ref. [7]. A measurement of $a_0(z=2)/a_0(0)$ to $\sim 20\%$ precision would distinguish between the three groups (constant, moderate, strong evolution); $\sim 10\%$ precision would separate the causal diamond from the empirical formula. This is the primary falsifiable test of the causal diamond program.

References

- [1] F. Lelli, S. S. McGaugh, J. M. Schombert, *Astrophys. J.* 816, L14 (2016).
- [2] M. Milgrom, *Astrophys. J.* 270, 365 (1983).
- [3] A. C. Mayer, A. F. Teklu, K. Dolag, R.-S. Remus, *Mon. Not. R. Astron. Soc.* 518, 257 (2023). arXiv:2206.04333.
- [4] M. Milgrom, "Cosmological variation of the MOND constant," *Phys. Rev. D* 91, 044009 (2015).
- [5] D. Nguyen, "The Geometric Origin of the MOND Acceleration Constant: The Causal Diamond Waist in Λ CDM," Zenodo (2026).
- [6] D. Nguyen, "Deriving the MOND Interpolation Function: Area-Law vs Volume-Law Entropy in the Causal Diamond," Zenodo (2026).
- [7] D. Nguyen, "Erratum: The Physical Event Horizon as the Correct Entropy Normalization," Zenodo (2026).
- [8] D. Nguyen, "An Empirical Cosmological Formula for the MOND Acceleration Constant," Zenodo (2026).
- [9] Planck Collaboration, Aghanim, N., et al., *Astron. Astrophys.* 641, A6 (2020).
- [10] SKA Cosmology Science Working Group, *Publ. Astron. Soc. Austral.* 37, e007 (2020).
- [11] S. Price et al., *Astron. Astrophys.* (2025). See also Rizzo et al., *Nature* 584, 201 (2020) for resolved kinematics at $z \sim 4$.
- [12] E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990).